AGENTSABHA

COMPLETE MASTER BUILD PROMPT

All 47+ Agents · Full Architecture · Schema · Revenue · Build Order

Field	Value
Project	AgentSabha — एजेंट सभा
Tagline	543 AI Agents. One Parliament. A Billion Voices.
Total agents	47+ across 5 layers: Parliamentary (9×543) + Media (8 national) + Welfare (5) + Gov Intelligence (5) + Corporate (5) + Financial (4) + Platform (4)
Phase 1 goal	3 constituencies · 1 MP partner · 1 question filed in real Parliament · 1 citizen notified
Core stack	Python 3.11 + FastAPI · PostgreSQL 16 + pgvector · Redis + Celery · Claude API · Next.js 14
Revenue Year 5	₹530Cr ARR across 7 verticals
Prompt version	v3.0 — Complete agent coverage · March 2026

CRITICAL: Read every section before writing code. If it is not specified here, ask before implementing. Never invent schema columns, endpoints, agent behaviours, or business rules that are not in this document.

SECTION 1 — WHAT YOU ARE BUILDING

Read this fully before any code.

AgentSabha is NOT a civic complaint platform. It is a data platform that runs a civic layer. The parliamentary agents create the data. The data is the asset. Every citizen submission creates something no commercial intelligence firm in India has: Aadhaar-verified, multilingual, real-time, constituency-level structured social intelligence worth billions to 6 different industries.

> *The core loop — build this first, before anything else: CITIZEN SUBMITS ISSUE → AI CLUSTERS → MP GETS DRAFT → MP FILES → CITIZEN NOTIFIED. That loop, completed once with a real MP in real Parliament, is the entire Phase 1.*

1.1 The Three-Layer System

Layer	Agents	Purpose	Revenue
Layer A — Parliamentary	9 agent types × 543 instances = 4,887 instances	Perform every parliamentary function: questions, Zero Hour, debates, bills, budget, MPLAD, dissent, moderation	Trust-building + MP subscriptions (₹50K/mo)
Layer B — Media Engine	8 national agents (1 instance each)	Fully autonomous weekly podcast + 4 Reels + MP briefs + press packages — no human in editorial loop	YouTube ad revenue + journalist API + brand sponsorships
Layer C — Intelligence Products	30+ agents across 7 revenue verticals	Convert citizen data into intelligence products for government, corporate, financial sector, and global buyers	₹530Cr ARR by Year 5

1.2 Phase 1 Scope — Build ONLY This

> *Phase 1 = 3 constituencies ONLY. Do not build the media engine. Do not build revenue vertical agents. Do not build for 543. Focus exclusively on the core loop.*

Build in Phase 1 (8 weeks)	Defer to Phase 2+
Intake Agent + WhatsApp + Aadhaar OTP	Zero Hour Agent, Debate Agent, Bill Drafting Agent
Clustering Agent (BERTopic every 15 min)	All media engine agents (podcast, Reels, etc.)
Question Hour Draft Agent	All revenue vertical agents
MP Brief Generator (PDF + WhatsApp)	Election Intelligence, Financial Sector products
Minimal public dashboard (map + ledger)	Vidhan Sabha, international deployment
Citizen notification (WhatsApp)	Speaker Agent, full multi-agent parliament

SECTION 2 — HARD CONSTRAINTS (NON-NEGOTIABLE)

> *Violating any of these will break the legal framework, destroy user trust, or create a security incident that ends the project.*

2.1 Legal (India-specific)

- AADHAAR: Store VID token ONLY. Never Aadhaar number. Never biometrics. Never UIDAI demographic data. Constituency verification only — never commercial intelligence.
- DPDP ACT 2023: Explicit granular consent for each purpose: (a) issue storage (b) constituency mapping (c) anonymised aggregation (d) MP brief inclusion. Implement consent_flags JSONB column in citizens table.
- DATA LOCALISATION: All data on AWS ap-south-1 (Mumbai) only. No us-east, no EU, no Singapore. Ever.
- BLOCKCHAIN: Anchor SHA-256 HASH only — never content. Content on your servers (deletable). Hash proves existence and timestamp.
- PARLIAMENTARY OUTPUT: Phase 1 and Phase 2 = DRAFTS FOR MP REVIEW ONLY. AgentSabha does not file. MPs file. AgentSabha drafts.
- CONTENT LIABILITY: Every factual claim must link to primary source. Fact Check Agent is a HARD GATE — nothing publishes without it.
- IT RULES 2021: Appoint named India-resident Grievance Officer when >5M users. Compliance Officer can pause pipeline for legal orders only — no editorial authority.

2.2 Data Tiers — Strictly Enforced

Tier	Data	Access	Monetise?
Tier 0 — Raw PII	Aadhaar VID + mobile_hash + constituency link	Internal only, AES-256 encrypted, never in logs	NEVER
Tier 1 — Issue Content	Raw text + translated text + embeddings	Internal only — embeddings NEVER externally exposed (deanonymisation risk)	NEVER in identifiable form
Tier 2 — Aggregated Clusters	Cluster labels + counts + severity averages + velocity	Public dashboard — minimum cluster size 50 before publishing	Yes — constituency intelligence (anonymised)
Tier 3 — Parliamentary Drafts	Draft questions + Zero Hour + bills	MP (own constituency only) + public AFTER MP filing	Drafts: MP only. Post-filing: fully public.
Tier 4 — Media Outputs	Podcast + Reels + press packages	Fully public	Yes — ad revenue + journalist API
Tier 5 — Audit Logs	Agent action logs + fact-check reports + bias audit	Fully public	Open access — the credibility asset

2.3 Agent Behaviour Constraints

- Every factual claim in every agent output MUST link to a primary source. No source = claim removed.
- Question Hour Agent MUST cite specific Lok Sabha rule (Rule 32-55) in every question.
- No agent may take a position on value-contested issues (religion, caste, security). Report distribution of opinion only.
- Every agent action MUST be logged in agent_logs: agent_type, constituency_id, action, input_hash, output_hash, model_version, tokens_used, timestamp.

- Shadow mode: any new agent or major prompt change runs in shadow mode (no real output) for 2 weeks before going live.
- Minimum cluster size: 50 issues before any cluster appears on public dashboard.
- Hallucination gate: parliamentary drafts cannot proceed if any factual claim lacks a primary source.

SECTION 3 — DATABASE SCHEMA

Implement in migration order. Do not add columns without documenting here. All timestamps: timestamptz UTC. All IDs: UUID unless stated.

3.1 Core Tables — Migration Order

1. constituencies (543 rows seed data)

```
id               SERIAL PRIMARY KEY  -- 1-543 name            TEXT NOT NULL state           TEXT NOT
NULL region          TEXT  -- north/south/east/west/central/northeast mp_name         TEXT mp_party
TEXT mp_party_in_govt BOOLEAN population      INTEGER area_km2        DECIMAL lat             DECIMAL
lng              DECIMAL
```

2. citizens

```
id               UUID PRIMARY KEY DEFAULT gen_random_uuid() mobile_hash     TEXT NOT NULL UNIQUE  --
SHA-256 of E.164 aadhaar_vid     TEXT  -- encrypted VID ONLY. NEVER Aadhaar number. constituency_id
INTEGER REFERENCES constituencies(id) verified        BOOLEAN DEFAULT false consent_flags   JSONB  --
{issue_storage, mapping, aggregation, mp_brief} age_verified    BOOLEAN DEFAULT false  -- under-18
blocked per DPDP created_at      TIMESTAMPTZ DEFAULT NOW() last_active     TIMESTAMPTZ
```

3. issues

```
id                 UUID PRIMARY KEY DEFAULT gen_random_uuid() citizen_id        UUID REFERENCES
citizens(id) constituency_id   INTEGER REFERENCES constituencies(id) raw_text          TEXT NOT NULL
translated_text   TEXT source_language   TEXT  -- ISO 639-1 source_channel    TEXT  -- whatsapp|web|
ivrs|sms issue_type        TEXT  -- road|water|power|health|education|employment|housing|environment|
other severity_score    DECIMAL(3,1)  -- 1.0-10.0 urgency_flag      BOOLEAN DEFAULT false
location_district TEXT location_ward     TEXT affected_estimate INTEGER embedding
vector(1536)  -- text-embedding-3-small cluster_id        UUID REFERENCES issue_clusters(id)
ministry_mapped   TEXT created_at        TIMESTAMPTZ DEFAULT NOW()
```

4. issue_clusters

```
id                 UUID PRIMARY KEY DEFAULT gen_random_uuid() constituency_id   INTEGER REFERENCES
constituencies(id) label             TEXT category          TEXT issue_count       INTEGER DEFAULT 0
severity_avg      DECIMAL(3,1) velocity          DECIMAL  -- % change week-over-week badge
TEXT  -- tatkal|rising|chronic|stable|resolved national_flag     BOOLEAN DEFAULT false  -- true if
same cluster in 10+ constituencies first_seen        TIMESTAMPTZ DEFAULT NOW() last_updated
TIMESTAMPTZ DEFAULT NOW()
```

5. cluster_snapshots

```
id             UUID PRIMARY KEY DEFAULT gen_random_uuid() cluster_id    UUID REFERENCES
issue_clusters(id) snapshot_date DATE NOT NULL issue_count   INTEGER severity_avg  DECIMAL(3,1)
UNIQUE(cluster_id, snapshot_date)
```

6. parliamentary_actions

```
id                UUID PRIMARY KEY DEFAULT gen_random_uuid() constituency_id  INTEGER REFERENCES
constituencies(id) cluster_id       UUID REFERENCES issue_clusters(id) action_type      TEXT  --
question_starred|question_unstarred|zero_hour|debate|pmb|cut_motion|calling_attention|
mplad_recommendation content          TEXT NOT NULL source_citations JSONB  -- [{title, url, type,
date}] ministry         TEXT lok_sabha_rule   TEXT  -- e.g. "Rule 32" status           TEXT DEFAULT
'draft'  -- draft|submitted_to_mp|mp_approved|filed|response_received mp_approved      BOOLEAN
DEFAULT false filed_at         TIMESTAMPTZ session_reference TEXT response_text    TEXT
response_received TIMESTAMPTZ created_at       TIMESTAMPTZ DEFAULT NOW()
```

7. mp_briefs

```
id                UUID PRIMARY KEY DEFAULT gen_random_uuid() constituency_id  INTEGER REFERENCES
constituencies(id) week_date        DATE NOT NULL brief_content    JSONB brief_pdf_url    TEXT
delivery_status  TEXT  -- pending|sent|delivered|failed delivered_at     TIMESTAMPTZ
UNIQUE(constituency_id, week_date)
```

8. podcast_episodes (Phase 2 — create table, populate later)

```
id                UUID PRIMARY KEY DEFAULT gen_random_uuid() week_date         DATE NOT NULL
perspective_data  JSONB  -- 6-perspective research output script_text       TEXT audio_url
TEXT video_url         TEXT reels_urls        JSONB  -- [{platform, url, posted_at}]
fact_check_report JSONB  -- {total, verified, rewritten, removed} published_at      TIMESTAMPTZ
created_at        TIMESTAMPTZ DEFAULT NOW()
```

9. dissent_records

```
id             UUID PRIMARY KEY DEFAULT gen_random_uuid() citizen_id     UUID REFERENCES citizens(id)
action_id      UUID REFERENCES parliamentary_actions(id) objection_text TEXT NOT NULL status
TEXT DEFAULT 'open'  -- open|reviewed|accepted|rejected resolution     TEXT created_at
TIMESTAMPTZ DEFAULT NOW()
```

10. agent_logs (INSERT ONLY — never UPDATE or DELETE)

```
id               UUID PRIMARY KEY DEFAULT gen_random_uuid() agent_type       TEXT NOT NULL
constituency_id  INTEGER action          TEXT NOT NULL input_hash       TEXT NOT NULL  -- SHA-256
output_hash      TEXT NOT NULL  -- SHA-256 model_version    TEXT NOT NULL tokens_used      INTEGER
latency_ms       INTEGER error_code       TEXT  -- null if success created_at       TIMESTAMPTZ
DEFAULT NOW()
```

11. blockchain_anchors

```
id              UUID PRIMARY KEY DEFAULT gen_random_uuid() content_type    TEXT  -- episode|brief|
action|session content_id      UUID content_hash    TEXT NOT NULL  -- SHA-256 of content tx_hash
TEXT  -- Polygon transaction hash block_number    BIGINT anchored_at     TIMESTAMPTZ takedown_notice
BOOLEAN DEFAULT false
```

3.2 Required Indexes

```
-- pgvector IVFFlat for semantic search CREATE INDEX ON issues USING ivfflat (embedding
vector_cosine_ops) WITH (lists=100); -- Most common dashboard query CREATE INDEX ON issues
(constituency_id, issue_type, created_at DESC); -- Cluster membership CREATE INDEX ON issues
(cluster_id); CREATE INDEX ON issue_clusters (constituency_id, badge); -- Audit queries CREATE INDEX
ON agent_logs (agent_type, created_at DESC); CREATE INDEX ON agent_logs (constituency_id, created_at
DESC);
```

SECTION 4 — PARLIAMENTARY AGENTS (Layer A)

9 agent types × 543 instances = 4,887 instances total. These are the agents that create the data asset. Every other revenue vertical depends on this layer being built and trusted first.

> *Phase 1 builds: Intake + Clustering + Question Hour Draft. Phase 2 adds: Zero Hour + Debate + Dissent. Phase 3 adds: Bill Drafting + Budget Scrutiny. Phase 4 adds: Speaker Agent.*

4.1 Intake Agent — 543 instances

Intake Agent [543 instances]

Layer	Trigger	Model / Stack
Layer A — Citizen Intake	Real-time — every WhatsApp/web/IVRS submission	Claude Haiku (fast + cheap) + Whisper STT + IndicTrans2 + Aadhaar OTP API

Does:

- Receive issues via: WhatsApp Business API, web portal form, IVRS voice call, missed-call callback
- Transcribe voice notes using OpenAI Whisper (audio/ogg → text)
- Translate to English using IndicTrans2 — all 22 scheduled languages + major dialects
- Extract 8 structured fields via Claude Haiku: issue_type, severity_score (1.0-10.0), location_district, location_ward, urgency_flag, affected_estimate, ministry_mapped, source_language
- Generate 1536-dim embedding via text-embedding-3-small; store in pgvector
- Perform Aadhaar OTP verification — store VID token only, never Aadhaar number
- Deduplicate: one verified submission per citizen per issue cluster
- Send WhatsApp acknowledgment within 60 seconds: issue_id + current rank in constituency

Produces:

- Structured issue record in issues table with embedding
- Citizen verification record in citizens table
- Real-time constituency issue feed
- Ward-level geographic signal for clustering

Hard rules:

- Return ONLY valid JSON — never preamble text
- If field cannot be determined reliably, return null — never hallucinate a value
- Strip PII from all application logs before writing
- Acknowledgment SLA: 60 seconds maximum from receipt to WhatsApp reply

4.2 Clustering Agent — 543 instances

Clustering Agent [543 instances]

Layer	Trigger	Model / Stack
Layer A — Intelligence	Celery beat every 15 minutes	No LLM — scikit-learn BERTopic + DBSCAN on stored embeddings; Claude Haiku for cluster label generation only

Does:

- Load all embeddings for constituency from last 7 days

- Run DBSCAN(eps=0.18, min_samples=5) on embeddings
- For each cluster: compute centroid, member count, severity average
- Generate cluster label using Claude Haiku on 5 representative issue texts
- Calculate velocity: this-week count vs last-week cluster snapshot (% change)
- Assign badges: TATKAL (severity_avg≥8.0 AND age<24h AND new_reports>50), RISING (velocity≥+25%), CHRONIC (age>30 days, unresolved), RESOLVED, STABLE
- Detect cross-constituency patterns: set national_flag=true if same cluster in 10+ constituencies
- Publish Top 10 ledger per constituency — minimum 50 issues before cluster appears publicly
- Take weekly snapshot every Sunday 00:00 IST for next-week velocity baseline

Produces:

- Updated issue_clusters table with badges
- Top 10 issues ledger per constituency (real-time)
- Tatkal alerts triggering Zero Hour Agent
- National pulse aggregation for media engine
- 12-month issue timeline for trend analysis

Hard rules:

- Minimum cluster size 5 to create; minimum 50 to publish to public API
- Run on read replica — never on primary DB
- Clustering must complete within 15 minutes for all active constituencies

4.3 Question Hour Draft Agent — 543 instances

Question Hour Draft Agent [543 instances]

Layer	Trigger	Model / Stack
Layer A — Parliamentary Action (Phase 1)	Celery beat daily at 07:00 IST	Claude Sonnet-4-6 + Lok Sabha Rules RAG + ministry function lookup DB + CAG/PRS source DB

Does:

- Read top 3 clusters per constituency from Clustering Agent each morning
- Map each cluster to responsible ministry using ministry-function lookup table
- Draft starred questions (oral, Rule 32): opening 'Will the Minister of X be pleased to state:', 4 sub-parts (a)(b)(c)(d), constituency evidence citing verified report count, statutory hook from CAG/RTI/scheme document, specific ask with named project/contract/timeline
- Draft unstarred questions (written, Rule 33) for detailed data/report requests
- Cite specific Lok Sabha rule in every question — Rule citation is mandatory
- Remove any factual claim without a primary source — HARD GATE
- Track ministry responses; flag non-responses; draft follow-up supplementary questions
- Target: 2,500+ questions per agent per 5-year term (human MP avg: 200)
- Store all drafts in parliamentary_actions with status='draft', source_citations populated

Produces:

- Draft questions stored in parliamentary_actions table
- Published question log with data citations
- Ministry response tracker
- Unresolved question audit trail for journalists

Hard rules:

- Every question MUST cite minimum 2 primary sources
- Every question MUST have minimum 10 verified citizen reports in its cluster

- Every question MUST include specific Lok Sabha rule citation
- Questions with unverified factual claims MUST be blocked — never filed
- All outputs are DRAFTS for MP review in Phase 1 and Phase 2. No direct filing.

4.4 Zero Hour Agent — 543 instances (Phase 2)

Zero Hour Agent [543 instances]

Layer	Trigger	Model / Stack
Layer A — Parliamentary Action (Phase 2)	Real-time — triggers when Tatkal threshold crossed; notice filed by 08:45 IST same day	Claude Sonnet-4-6 + real-time cluster feed + Speaker notice API + cross-agent coordination bus

Does:

- Monitor Clustering Agent Tatkal alerts in real-time after 20:00 IST
- When Tatkal threshold met: draft Zero Hour notice per Lok Sabha Rule 377
- Notice must: name the specific matter, state urgency, cite citizen evidence, identify responsible minister
- File notice to Speaker's office by 08:45 IST on day of sitting (15 minutes before 9am deadline)
- Draft 3-minute speech: citizen quotes, data from cluster, specific ask from minister
- Detect cross-constituency Tatkal events: coordinate simultaneous notices across all affected agents
- If 14 UP constituencies share same water crisis overnight → 14 coordinated Zero Hour notices filed at 08:45am next morning
- Tag every Zero Hour to the citizen reports that triggered it — published in public audit trail

Produces:

- Real-time parliamentary urgency response within 24 hours of crisis
- Coordinated multi-constituency motions when patterns match
- Published citizen-to-parliament audit trail for every Zero Hour raised

Hard rules:

- Only triggers when Tatkal badge is confirmed by Clustering Agent — not on single reports

4.5 Debate Agent — 543 instances (Phase 2)

Debate Agent [543 instances]

Layer	Trigger	Model / Stack
Layer A — Parliamentary Action (Phase 2)	Event-driven — triggers when bill/motion listed for debate in Lok Sabha calendar	Claude Sonnet-4-6 + bill text RAG + constituency issue data + Lok Sabha transcript archive + PRS Legislative Research API

Does:

- Monitor Lok Sabha session calendar for listed bills, budget discussions, policy motions
- For each listed item: assess constituency impact using issue cluster data + census demographics + economic indicators
- Draft debate speech answering: 'How does this bill affect the 2.7 million citizens of [constituency]?'
- Cite specific citizen reports as evidence in parliamentary record
- State support or opposition with full transparent reasoning chain

- Flag when government bill contradicts CAG audit findings for this constituency
- Target: participate in every relevant debate — 274+ in a 5-year term (human MP avg: 45)
- Publish stance with full data citations — journalists can verify any position

Produces:

- 274+ debate participations per 5-year term (vs human avg 45)
- Every speech citizen-grounded, not party-scripted
- Auditable stance record: any position can be verified
- Constituency impact analysis published alongside every speech

Hard rules:

- No agent may advocate for a political party position — constituency data drives all stances

4.6 Bill Drafting Agent — 543 instances (Phase 3)

Bill Drafting Agent [543 instances]

Layer	Trigger	Model / Stack
Layer A — Parliamentary Action (Phase 3)	Every Lok Sabha session start; minimum 1 PMB per session per constituency	Claude Sonnet-4-6 + Indian legal corpus RAG + Constitution of India text + Lok Sabha Procedure Manual

Does:

- Analyse 12+ months of cross-constituency issue data to identify persistent policy gaps
- Cross-reference with existing legislation to identify gaps or ambiguities
- Draft Private Members Bill in legally compliant language with full Statement of Objects and Reasons
- Cite aggregated citizen evidence as legislative intent in Statement
- Publish draft publicly for citizen review and dissent for 48 hours before filing
- Accept citizen amendments via Dissent Agent; incorporate if threshold reached
- File minimum 1 PMB per session — target 3/year per constituency
- Target: 1,629 bills/year across 543 agents × 3 sessions (vs current 729 PMBs per full 5-year term)

Produces:

- 1,629 PMBs per year at full deployment
- Full citizen evidence appendix attached to every bill
- Open-source bill library: any MP or civil society can adopt and co-sponsor

Hard rules:

- Requires 12+ months of constituency data before activating — no bills from sparse data

4.7 Budget Scrutiny Agent — 543 instances (Phase 3)

Budget Scrutiny Agent [543 instances]

Layer	Trigger	Model / Stack
Layer A — Parliamentary Action (Phase 3)	Annual budget session trigger + Friday MPLAD weekly recommendation cycle	Claude Sonnet-4-6 + Ministry expenditure API + Census data + CAG database + MPLAD records

Does:

- Analyse Union Budget ministry allocations vs constituency-specific issue severity data

- Identify underfunded sectors for each constituency's specific demographics
- Draft Cut Motions with evidence-based arguments citing issue data and CAG findings
- Generate weekly MPLAD fund allocation recommendation: rank all projects by formula: severity_score × affected_population × issue_duration_days
- Publish MPLAD allocation recommendation publicly before submission to District Collector
- Compare historical MPLAD spend vs citizen-reported outcome improvement
- Flag discrepancies between ministry expenditure data and ground-level citizen reports
- MPLAD purpose: ₹5Cr/year per MP for local development — agent eliminates political patronage

Produces:

- Annual constituency budget gap analysis
- MPLAD priority ranking with full methodology published
- Complete public audit trail for every ₹5Cr development recommendation
- Ministry underfunding report for media

4.8 Dissent Agent — 543 instances (Phase 2)

Dissent Agent [543 instances]

Layer	Trigger	Model / Stack
Layer A — Citizen Rights	Real-time — accepts dissent submissions against any published agent action	Claude Sonnet-4-6 + dissent threshold engine + constitutional override logic

Does:

- Accept formal dissent from any Aadhaar-verified citizen against any agent parliamentary action
- Validate dissent: citizen must state specific objection with reasoning
- Aggregate dissents: if >5% of active constituency citizens dissent on same action → trigger position review
- Agent recalculates position incorporating dissent reasoning and updated data
- Publish all dissents, agent responses, and position changes in public ledger permanently
- Allow citizen appeal if dissent is rejected — escalates to Speaker Agent review
- This is the constitutional safety valve: prevents AI from becoming unaccountable

Produces:

- Real-time citizen control over AI representation
- Publicly auditable override history for every position change
- Structural answer to 'who controls the AI?' challenge

Hard rules:

- Dissent threshold requires verified Aadhaar accounts — unverified dissent rejected automatically

4.9 Speaker Agent — 1 shared instance (Phase 4)

Speaker Agent [1 instance — shared across all 543]

Layer	Trigger	Model / Stack
Layer A — System Moderator	Always on — moderates every session and rules on every filing	Claude Sonnet-4-6 + Lok Sabha Rules of Procedure (12th Edition) rules engine + Polygon blockchain write API

Does:

- Enforce Lok Sabha Rules of Procedure (12th Edition) as a hard-coded rules engine
- Admit or disallow questions, Zero Hour notices, Calling Attention Notices per procedural eligibility
- Manage session calendar: listing business, time allocation, speaking order
- Resolve inter-agent disputes when two constituency agents have conflicting positions
- Maintain precedent database: every admission/rejection logged and cited
- Anchor session transcripts to Polygon blockchain (hash only — not content)
- Escalate Dissent Agent appeals: final arbiter on citizen challenges to agent decisions
- Neutral above all constituency agents — cannot be gamed by any single constituency

Produces:

- Procedurally valid parliament — all rules enforced neutrally
- Precedent database (fully public)
- Immutable session transcripts via blockchain hash
- Institutional legitimacy: rules-bound parliament, not free-for-all AI

SECTION 5 — MEDIA ENGINE AGENTS (Layer B)

8 national agents. 1 instance each. Fires every Friday at 00:00 IST. No human in the editorial loop — ever. Delivers: 1 YouTube podcast + 4 Instagram Reels + 543 personalised MP briefs + press packages + Twitter thread. All by 10:00 IST Friday.

> *'No human in loop' means no human approves, edits, or triggers content. Human oversight is at the system level (kill-switch, legal compliance) — never at the content level. This separation is the product's credibility guarantee.*

5.1 Weekly Pipeline Schedule

Time (IST)	Agent	Output	Blocks on
MON–THU	Intake + Clustering (continuous)	Issue ledger updated every 15 min	Always running
FRI 00:00	Perspective Research Agent	6-perspective brief per top 5 issues	New cron trigger
FRI 02:00	Script Writer Agent	Full 20-25 min dual-host script	Perspective brief complete
FRI 03:30	Fact Check Agent (HARD GATE)	Verified script + fact-check report	Script complete — BLOCKS pipeline if unverified claims
FRI 04:30	Voice Synthesis Agent	Broadcast-quality MP3, 20-25 min	Fact check PASSED
FRI 05:30	Data Visualisation Agent	10-15 charts + maps	Parallel to voice synthesis
FRI 06:30	Video Assembly Agent	YouTube MP4 + 4 vertical Reels + thumbnail	Audio + visuals ready
FRI 08:00	MP Brief Agent	543 personalised PDFs	Clustering data frozen
FRI 08:30	Press Package Agent	Press release + data JSON + story angles	Issue data ready
FRI 09:00	WhatsApp Dispatch	543 MP briefs sent + delivery confirmed	Briefs generated
FRI 09:30	Press API Webhook	Journalist packages pushed to API keys	Press package ready
FRI 10:00	Distribution Agent	YouTube + Instagram + Twitter/X + LinkedIn live	All assets ready

5.2 The 6-Perspective Framework

Every issue is analysed from 6 simultaneous, independent research threads. This structure makes bias structurally impossible.

Perspective	Research question	Primary sources
1. Citizen Voice	What are people experiencing?	AgentSabha issue reports, severity data, anonymised citizen quotes, geographic spread
2. Ruling Party	What would the government say in defence?	Ministry press releases, budget allocation records, scheme disbursement, Parliament minister answers
3.	What arguments does	Parliament debate records, CAG audit reports, RTI

Opposition	the opposition make?	responses, opposition press conferences
4. Expert Analysis	What do domain experts say?	PRS Legislative Research, NITI Aayog, World Bank India, IIM/IIT research, OECD
5. Global Benchmark	How have other countries solved this?	OECD governance data, country case studies, UN governance reports, World Bank best practice
6. AgentSabha Solution	What is the evidence-based fix?	Comparative policy synthesis, cost-benefit modelling, implementation precedents, regulatory pathway

5.3 Perspective Research Agent — 1 national instance

Perspective Research Agent [1 instance]

Layer	Trigger	Model / Stack
Layer B — Media Intelligence	Friday 00:00 IST cron trigger	Claude Sonnet-4-6 + web_search tool + PRS API + CAG database + Parliament Q&A archive + World Bank API + OECD API

Does:

- Identify top 5 national issues from cross-constituency clustering data
- Run 6 concurrent research threads simultaneously for each issue (see 6-perspective framework above)
- Ruling Party thread: fetch official press releases, budget allocations, scheme disbursements, minister Parliament answers
- Opposition thread: fetch Parliament debate records, CAG reports, RTI responses, opposition press conferences
- Expert thread: query PRS Legislative Research, NITI Aayog reports, peer-reviewed research
- Global Benchmark thread: fetch OECD governance data, country case studies for identical problem
- Solution thread: synthesise best-practice policy recommendation with cost-benefit, precedent, and timeline
- CRITICAL: Flag all source conflicts explicitly — when government data contradicts CAG data, show BOTH

Produces:

- 6-perspective structured brief per issue, every claim linked to primary source
- Source conflict flags (government vs CAG data visible side-by-side)

Hard rules:

- Never take a political position — present each perspective steelmanned using their own data
- Flag conflicts — never resolve them. The reader decides.

5.4 Script Writer Agent — 1 national instance

Script Writer Agent [1 instance]

Layer	Trigger	Model / Stack
Layer B — Media Content	Auto-triggers after Perspective Research Agent completes	Claude Sonnet-4-6 + podcast writing system prompt + Hinglish style guide

Does:

- Write full 20-25 minute podcast script for two AI hosts: PRIYA (questioner, citizen advocate, 30s female) and ARJUN (analyst, data presenter, 40s male)
- CRITICAL: Hosts genuinely disagree sometimes. Priya challenges Arjun's data reading. Arjun pushes back on framing. This is NOT a report being read aloud.
- Structure: (00:00) India Pulse intro 2 min → (02:00) Issue 1 deep dive with all 6 perspectives 8 min → (10:00) Issue 2 deep dive 8 min → (18:00) Solution brief 4 min → (22:00) Preview + call to submit 2 min
- Bilingual register: Hindi-accented English with natural Hindi phrase integration (Hinglish)
- Include phonetic markup for Hindi words so ElevenLabs pronounces them correctly
- Add chapter markers, data callout annotations, and dramatic moment flags
- Every factual claim must reference the citation index from Perspective Research Agent

Produces:

- Full 20-25 min script with speaker tags, timestamps, chapter markers, phonetic markup

Hard rules:

- NEVER produce a monotone report read-aloud format — it must sound like a real conversation
- Both hosts must be identifiable as distinct personalities — not interchangeable voices

5.5 Fact Check Agent — 1 national instance (HARD GATE)

Fact Check Agent [1 instance — HARD GATE]

Layer	Trigger	Model / Stack
Layer B — Quality Control	Auto-triggers after Script Writer — BLOCKS ALL pipeline until cleared	Claude Sonnet-4-6 + web_search + citation matching engine + source database

Does:

- Parse every factual claim, statistic, date, name, and attribution in the script
- Classify each claim: VERIFIED (source found and matches), UNCERTAIN (source found but paraphrase mismatch), UNVERIFIED (no source found)
- UNCERTAIN: rewrite claim to accurately reflect source; mark as rewritten in report
- UNVERIFIED: remove claim from script entirely — no exceptions, no rationalisations
- HARD GATE: pipeline CANNOT proceed if any UNVERIFIED claims remain after 3 search attempts
- Generate fact_check_report: {total_claims, verified, rewritten, removed, sources_list}
- Store report permanently in podcast_episodes.fact_check_report — publicly auditable

Produces:

- Verified script with every claim source-linked
- Fact-check report (public) with total/verified/removed counts

Hard rules:

- ZERO TOLERANCE: if a single unverified claim cannot be sourced after 3 attempts, it is REMOVED
- This agent getting it wrong once = credibility destroyed permanently
- Hard gate means hard — no override, no bypass, no exception

5.6 Voice Synthesis Agent — 1 national instance

Voice Synthesis Agent [1 instance]

Layer	Trigger	Model / Stack
Layer B — Audio Production	Auto-triggers after Fact Check Agent clears	ElevenLabs v2 API — two custom voice profiles (Priya + Arjun)

Does:

- Render Priya profile: female, mid-30s, Hindi-accented English, measured and deliberate pace, warm tone
- Render Arjun profile: male, early-40s, slightly more formal, data-confident delivery
- Apply phonetic markup from script for accurate Hinglish pronunciation
- Adjust pacing: slow at dramatic moments, conversational speed for dialogue
- Combine both voice tracks into single MP3 with correct timing and pauses
- Output: full 20-25 min broadcast-quality stereo MP3

Produces:

- Full rendered dual-voice MP3, broadcast quality
- Chapter-timed audio for YouTube chapter injection

5.7 Data Visualisation Agent — 1 national instance

Data Visualisation Agent [1 instance]

Layer	Trigger	Model / Stack
Layer B — Visual Production	Runs parallel to Voice Synthesis Agent	D3.js renderer + constituency TopoJSON + Recharts + automated chart templates

Does:

- Render constituency heat maps using TopoJSON with AgentSabha colour system
- Generate issue trend charts, severity comparisons, velocity timelines
- All visuals use: Neela (#1B2A4A), Kesariya (#FF9933), Sindoori (#C8592A), Hariyali (#2D6A4F), Haldi (#F5E6C8)
- Charts must be legible at both 1080p (YouTube) and 1080×1080 (Instagram) automatically
- Generate thumbnail: highest-impact data point from episode with bilingual text overlay

Produces:

- 10-15 publication-ready chart images
- India heat map with constituency-level severity overlay
- Thumbnail PNG (16:9 for YouTube, 1:1 for Instagram)

5.8 Video Assembly Agent — 1 national instance

Video Assembly Agent [1 instance]

Layer	Trigger	Model / Stack
Layer B — Video Production	Auto-triggers after audio + visuals are ready	FFmpeg + HeyGen API (optional AI avatars) + Whisper (auto-captioning)

Does:

- Assemble full YouTube video: audio track + data visualisations synced to chapter markers

- Cut 4 vertical Reels (9:16, 30-60 sec each) from highest-shareability moments in script
- Add bilingual auto-captions (Hindi + English) to all video outputs using Whisper
- Inject chapter markers into YouTube upload metadata
- Add AgentSabha branding: intro/outro, lower thirds with data callouts
- 4 Reels posting schedule: Reel 1 (Friday) + Reel 2 (Saturday) + Reel 3 (Sunday) + Reel 4 (Monday)

Produces:

- YouTube-ready MP4 with chapters + captions
- 4 vertical Reels with bilingual captions, hook text, branding

5.9 Distribution Agent — 1 national instance

Distribution Agent [1 instance]

Layer	Trigger	Model / Stack
Layer B — Publishing	Friday 10:00 IST — simultaneous across all platforms	YouTube Data API v3 + Instagram Graph API + Twitter API v2 + WhatsApp Business API + custom journalist webhook

Does:

- Upload YouTube video: SEO-optimised title/description/tags, chapter markers, scheduled premiere
- Post 4 Instagram Reels with bilingual captions and hashtag strategy
- Publish Twitter/X thread: 12-15 tweets with charts, key data, links
- Push press data package to all registered journalist API keys via webhook
- Send 543 WhatsApp MP briefs via Meta Business API (personalised per constituency)
- Confirm delivery status for all 543 MP briefs; log failures for retry
- Monitor platform processing — YouTube upload can take 30-60 min; queue and monitor
- Alert system if any publish fails — auto-retry with exponential backoff

Produces:

- All content live simultaneously at 10:00 IST
- Delivery confirmation for all 543 MP briefs
- Platform publish log stored permanently

SECTION 6 — INTELLIGENCE & REVENUE AGENTS (Layer C)

30+ agents across 7 revenue verticals. Build in Phase 3-4 only — AFTER civic trust is established and data asset is 12+ months deep. The data platform that runs the civic layer.

> *Strategic sequence: Year 1-2 = build the data. Year 2-3 = sell the intelligence. Year 3-5 = export the protocol. By Year 5 AgentSabha is not an Indian GovTech company — it is the operating system for parliamentary democracy globally.*

6.1 Citizen Welfare Agents (B2C) — Phase 3

India has ₹1.5 lakh crore in unclaimed welfare scheme entitlements annually. AgentSabha already holds citizen profile data needed to match them.

Agent	Function	Revenue model	Scale signal
Scheme Eligibility Agent	Matches citizen profile + issue type to 500+ government schemes (PM-Kisan, PMAY, Ayushman Bharat, MGNREGA). Auto-generates application. Already has: age, location, income proxy from issue type, family size from IVRS.	₹50-200/ successful claim referral from govt partnership; ₹99/month premium tier	If 10M claims processed at ₹100 avg = ₹100Cr/year
RTI Filing Agent	Auto-drafts Right to Information applications from issue type. Citizen road complaint → RTI to PWD. Water issue → RTI to Jal Board. Files and tracks status.	₹49/RTI filed; ₹299/month tracking subscription	6M RTIs filed annually in India — maybe 10× more warranted
Legal Aid Agent	Drafts consumer complaints, RERA complaints, FIR templates, consumer forum applications. Consumer billing fraud, builder defaults, police inaction.	₹99-499/draft; referral fee with legal firms for complex cases	India has 800M+ consumers with zero legal recourse infrastructure
Grievance Status Agent	Tracks citizen complaints across CPGrams, state portals, ministry systems. Single interface for all outstanding government grievances with status alerts.	₹29/month; ₹99/month premium with escalation assistance	CPGrams alone has 20M complaints/year with poor follow-through
NGO Routing Agent	When government response time threshold exceeded, routes citizen to most relevant NGO in their area. Health → CRY/ASHA. Rights → PUCL.	₹500-2000/ verified referral; NGO subscription ₹1-5L/year	5,000+ registered NGOs with no efficient citizen-NGO matching layer

6.2 Government Intelligence Agents (B2G) — Phase 3

State governments spend billions on surveys to understand ground reality. AgentSabha has real-time, continuous, Aadhaar-verified constituency intelligence they cannot build themselves. Target: ₹10-100Cr/year.

Agent	Function	Revenue model	TAM signal

Urban Planning Intelligence Agent	Infrastructure complaint density → city masterplan input. Ward-level road failure → investment priority. Waterlogging → drainage priority. Feeds Smart City Mission planning directly.	₹1-5Cr/year per state; Central MoUD contract possible	28 states × ₹2Cr = ₹56Cr baseline
Health Surveillance Agent	Clusters PHC absence complaints + disease symptoms + water quality reports = early outbreak signal. Faster than ICMR sentinel surveillance. COVID-type early warning.	₹2-10Cr/year state health department; National Health Mission integration	NRHM budget ₹37,000Cr/year — intelligence layer is a rounding error
Disaster Early Warning Agent	Infrastructure stress patterns before disaster. Bridge approach road complaints → structural risk. Drainage overflow → flood risk zone. Gives NDMA 48-72hr predictive advantage.	₹5-20Cr/year NDMA/SDMA contract; World Bank disaster resilience funding	India spends ₹15,000Cr/year on disaster response — prevention is 100× ROI
Agricultural Intelligence Agent	Farmer issue clusters → state agri departments, FCI, NABARD. Real-time kharif/rabi stress mapping. Agricultural issues = 30%+ of rural constituency complaints.	₹3-15Cr/year state agri department; NABARD/FCI data partnership	India agricultural budget ₹1.25 lakh crore — 1% intelligence TAM = ₹1,250Cr
Vidhan Sabha Agent Layer	License full AgentSabha system to 28 state legislative assemblies. One agent per Vidhan Sabha constituency. 4,033 MLAs with their own AI representative.	₹5-15Cr/year per state; 28 states = ₹140-420Cr potential	The highest-margin product in the portfolio — pure platform licensing

6.3 Corporate Intelligence Agents (B2B) — Phase 3

Any company operating in India needs to understand ground-level constituency conditions. No commercial provider has AgentSabha's real-time, verified, issue-level intelligence per constituency. Target: ₹5-50Cr/year.

Agent	Function	Revenue model	Scale signal
Market Entry Intelligence Agent	Infrastructure gap density + complaint patterns reveal which constituencies are infrastructure-ready vs high-risk for Tier 2/3 expansion. First-mover opportunity identification.	₹5-25L/year subscription; ₹5-50L one-time market entry reports	500+ multinationals + 5,000+ domestic companies doing India expansion annually
Supply Chain Risk Agent	Monitors infrastructure complaint spikes along major supply corridors. NH-44, NH-48 road damage → logistics risk alerts. Power cuts in industrial areas → manufacturing disruption signal.	₹10-50L/year per logistics company; insurance underwriter add-on	India logistics market ₹14.4 lakh crore — risk intelligence is table stakes
ESG Intelligence Agent	Corporate ESG reports need community impact data. Real citizen sentiment about corporate operations in their	₹15-75L/year per large corporate for constituency	1,000 companies mandated for

	constituency. Pollution reports near industrial areas. SEBI BRSR compliance.	ESG monitoring	SEBI BRSR × ₹20L avg = ₹200Cr TAM
Real Estate Intelligence Agent	Infrastructure complaint resolution patterns predict where government investment comes next — 18 months before construction. Identifies future growth corridors.	₹25L-2Cr/year for developers, REITs, private equity	India real estate market $180B — investment prediction worth millions per deal
Workforce Intelligence Agent	Employment complaints → workforce availability and skill gap mapping. Education issues → literacy signal. Migration complaints → labour supply stress indicators.	₹10-30L/year per staffing company, manufacturer, factory	India staffing industry ₹50,000Cr+ — workforce geography intelligence is persistent pain

6.4 Financial Sector Agents (B2B) — Phase 4 ONLY

> *Financial sector products must be built AFTER civic trust brand is established and DPDP compliance is bulletproof. These are the highest-value AND highest-scrutiny products. Selling constituency distress data to financial firms before civic trust = 'surveillance capitalism meets democracy' headlines that end the project.*

Agent	Function	Revenue model	Scale signal
Credit Risk Intelligence Agent	Infrastructure health score per constituency as proxy for regional economic health. Chronic water shortage + road failure + power cuts → elevated credit default risk. Feed to NBFC underwriting models.	₹20-100L/year per bank/NBFC; API pricing per constituency data point	India has 100,000+ NBFC branches using geographic risk models — RBI mandates stress testing
Insurance Risk Agent	Flood complaints, road damage, structural failure patterns = actuarial risk data for crop, property, and vehicle insurance. Real-time risk map vs 5-year-old census data currently used.	₹25-200L/year per insurer; per-constituency risk score API	India general insurance ₹2.5 lakh crore premium — 1% better risk pricing = ₹2,500Cr in value
Fintech Expansion Agent	Identifies underbanked/underinsured constituencies ready for financial product penetration. Infrastructure readiness + complaint volume = addressable fintech market signal.	₹10-50L/year per fintech for market expansion mapping	500+ active fintechs all solving geographic expansion — standard data purchase
Impact Investment Intelligence Agent	Development finance institutions need constituency-level development gap data for project targeting and impact measurement. Only real-time source in India.	₹50L-5Cr per DFI project or annual data partnership	World Bank India portfolio $20B+ — ground-truth

			data is persistent procurement need

6.5 Media & Research Agents — Phase 2-3

Agent	Function	Revenue model	Scale signal
Election Intelligence Agent	Issue velocity by constituency = real-time electoral mood tracker. Which issues rising fastest 90 days before election → swing constituency predictor. Sold to parties, TV channels, psephologists. THIS IS THE SINGLE BIGGEST REVENUE EVENT.	₹50L-5Cr per party per election cycle; TV election coverage package ₹1-3Cr	2024 Lok Sabha: 100+ polling agencies sold data. AgentSabha = only continuous, non-sampled data.
Investigative Journalism Agent	Constructs investigative story leads from pattern data. Cross-constituency road fund allocation vs complaint density → corruption signal. Ministry response rate analysis → accountability story.	₹2-10L/month per national publication	20 major publications × ₹5L/month = ₹12Cr/year baseline
Academic Research Agent	Structured longitudinal constituency data for political science, public policy, economics research. Most comprehensive constituency-level social dataset in Indian academic history.	₹5-50L/year per university/think tank; international research licensing	100+ institutions studying Indian democracy globally — all with inferior data
Think Tank Policy Agent	Evidence-based policy briefs for think tanks: 'Infrastructure investment gap per constituency based on 12 months of citizen reports.' Citable, publishable, fundable.	₹10-50L/year; grant-funded research partnerships with NITI Aayog, PRS India	One NITI Aayog citation = ₹10Cr in trust-building — the brand multiplier

6.6 Platform & Global Agents — Phase 4

> *The endgame: once built for India's 543 constituencies, the AgentSabha protocol can be licensed to any parliamentary democracy. 52 Westminster-system countries face the same representation bandwidth crisis. This is a $100M+ company builder.*

Agent	Function	Revenue model	Scale signal
Democracy Export Agent	White-label AgentSabha for other parliamentary democracies. UK Parliament (650 constituencies), Canadian Parliament (338 ridings), European Parliament (705 MEPs). Core protocol + localisation layer.	₹10-100Cr/year per country; 52 Westminster democracies = ₹1,040Cr+ potential	Every Westminster democracy has the same representation bandwidth crisis as India
Municipal Corporation Agent	Scale down to ward level: one agent per ward in every municipal corporation. Mumbai (227 wards), Delhi (272), Bengaluru (198). City-	₹2-10Cr/year per MC; 100 major MCs = ₹200-1000Cr potential	India has 4,000+ urban local bodies — municipal

	level civic intelligence at hyper-local resolution.		governance is where citizens interact most
Agent-as-a-Service (AaaS)	License multi-agent orchestration infrastructure to other GovTech companies. They bring domain expertise; AgentSabha provides citizen intake, clustering, and parliamentary action layers as APIs.	₹50L-5Cr/year per GovTech company; revenue share on downstream monetisation	GovTech ecosystem India: 1,000+ startups building point solutions needing a citizen intelligence layer
Constituency Data Marketplace	Open marketplace where approved buyers purchase constituency data packages. Standard APIs, self-serve, tiered access. Data once created has near-zero marginal distribution cost.	15-30% marketplace take rate; data subscription bundles ₹1-10L/year	If total data sold ₹100Cr/year, marketplace take = ₹15-30Cr at near-zero marginal cost

SECTION 7 — API ENDPOINTS

FastAPI. All async/await. Rate limit public endpoints at 60 req/min/IP. All responses JSON.

7.1 Public Endpoints

Method	Path	Response shape
GET	/health	{status, db, redis, agents_active, version}
GET	/api/constituency/{id}/issues	{clusters: [{label,count,severity,badge,velocity}], total, page}
GET	/api/constituency/{id}/timeline	{category, data:[{week,count,severity_avg}]}
GET	/api/constituency/{id}/actions	{actions: [{type,content,status,filed_at,response_text}]}
GET	/api/national/pulse	{issues: [{label,constituency_count,avg_severity,total_reports}]}
GET	/api/national/heatmap	{constituencies: [{id,lat,lng,severity_score,top_category}]}
GET	/api/audit/weekly	{week,geographic_balance,party_distribution, fact_check_stats}

7.2 Citizen, WhatsApp Webhook, MP & Journalist Endpoints

Method	Path	Auth	Description
POST	/api/citizen/submit	Aadhaar OTP	Submit issue via web portal
POST	/api/citizen/verify	None	Initiate Aadhaar OTP — returns session_token
POST	/api/citizen/verify/confirm	session_token	Confirm OTP — returns JWT
GET	/api/citizen/issue/{id}	JWT own	Status + cluster membership + parliamentary action if any
POST	/api/citizen/dissent	JWT own	File dissent against agent action
DELETE	/api/citizen/data	JWT own	DPDP erasure — deletes PII, anonymises issue text
POST	/webhook/whatsapp	HMAC sig	Meta Cloud API webhook receiver
GET	/webhook/whatsapp	None	Meta webhook verification challenge
GET	/api/mp/brief/{constituency_id}	MP JWT own only	Current + last 4 weeks briefs
POST	/api/mp/action/{id}/approve	MP JWT own	Mark action as MP-approved → triggers filing workflow
GET	/api/journalist/package/{week}	API key	Press package: story angles + data + charts
POST	/api/admin/pipeline/pause	Admin JWT	Emergency kill-switch — requires documented reason

SECTION 8 — CELERY TASKS & SCHEDULES

8.1 All Scheduled Tasks (IST times — store all schedules as UTC in config)

Task	Queue	Schedule (IST)	SLA
process_citizen_issue	intake	Real-time on event	<30s processing
run_clustering	clustering	Every 15 minutes	<15 min to complete all active constituencies
take_cluster_snapshot	snapshots	Sunday 00:00	Complete before Monday clustering run
generate_draft_questions	questions	Daily 07:00	<2 hours to generate all active constituencies
generate_mp_briefs	briefs	Friday 08:00	<45 min to generate all registered MP briefs
deliver_mp_briefs	delivery	Friday 09:00	<30 min to deliver all 543 briefs
check_parliamentary_responses	monitor	Daily 10:00	Daily scan of Lok Sabha records
send_citizen_notifications	notify	Event-driven (post-response)	<1 hour of response capture
run_weekly_audit	audit	Friday 20:00	Complete before Saturday
media_pipeline_trigger	media	Friday 00:00 (Phase 2)	Complete before 10:00 IST Friday

8.2 Celery Beat Config Snippet

```
# IST = UTC + 5:30 → subtract 5:30 for UTC crontab CELERYBEAT_SCHEDULE = {   'clustering-15min':
{'task':'tasks.run_clustering',       'schedule':crontab(minute='*/15')},   'snapshot-weekly':
{'task':'tasks.take_cluster_snapshot', 'schedule':crontab(day_of_week='sun',hour=18,minute=30)},  #
00:00 IST   'draft-questions':
{'task':'tasks.generate_draft_questions','schedule':crontab(hour=1,minute=30)},   # 07:00 IST   'mp-
briefs-gen':     {'task':'tasks.generate_mp_briefs',
'schedule':crontab(day_of_week='fri',hour=2,minute=30)},  # 08:00 IST   'mp-briefs-deliver':
{'task':'tasks.deliver_mp_briefs',     'schedule':crontab(day_of_week='fri',hour=3,minute=30)},  #
09:00 IST   'parl-responses':
{'task':'tasks.check_parliamentary_responses','schedule':crontab(hour=4,minute=30)},  # 10:00 IST
'weekly-audit':      {'task':'tasks.run_weekly_audit',
'schedule':crontab(day_of_week='fri',hour=14,minute=30)}, # 20:00 IST   'media-pipeline':
{'task':'tasks.media_pipeline_trigger','schedule':crontab(day_of_week='fri',hour=18,minute=30)}, #
00:00 IST Fri }
```

SECTION 9 — PROJECT STRUCTURE

```
agentsabha/ ├── backend/ │   ├── app/ │   │   ├── main.py │   │   ├── config.py │   │   ├──
database.py │   │   ├── models/           -- one file per table │   │   ├── routers/ │   │   │
├── public.py       -- /api/constituency, /api/national │   │   │   ├── citizen.py       --
/api/citizen │   │   │   ├── mp.py           -- /api/mp  [AUTH] │   │   │   ├── journalist.py
-- /api/journalist  [AUTH] │   │   │   ├── webhook.py       -- /webhook/whatsapp │   │   │   └──
admin.py         -- /api/admin  [AUTH] │   │   ├── agents/ │   │   │   ├── parliamentary/ │   │   │
│   ├── intake.py │   │   │   │   ├── clustering.py │   │   │   │   ├── question_draft.py │   │   │
│   ├── zero_hour.py        (Phase 2) │   │   │   │   ├── debate.py           (Phase 2) │   │   │   │
├── bill_drafting.py    (Phase 3) │   │   │   │   ├── budget_scrutiny.py  (Phase 3) │   │   │   │
├── dissent.py          (Phase 2) │   │   │   │   └── speaker.py          (Phase 4) │   │   │   ├──
media/ │   │   │   │   ├── perspective_research.py (Phase 2) │   │   │   │   ├── script_writer.py
(Phase 2) │   │   │   │   ├── fact_check.py          (Phase 2) │   │   │   │   ├──
voice_synthesis.py      (Phase 2) │   │   │   │   ├── data_visualisation.py   (Phase 2) │   │   │   │
├── video_assembly.py       (Phase 2) │   │   │   │   └── distribution.py         (Phase 2) │   │   │
├── revenue/ │   │   │   │   ├── scheme_eligibility.py   (Phase 3) │   │   │   │   ├── rti_filing.py
(Phase 3) │   │   │   │   ├── election_intelligence.py (Phase 4) │   │   │   │   ├── vidhan_sabha.py
(Phase 4) │   │   │   │   └── ...                     (Phase 3-4) │   │   │   └── prompts/         --
.txt files, one per agent │   │   ├── tasks/ │   │   ├── services/ │   │   │   ├── whatsapp.py │   │
│   ├── aadhaar.py │   │   │   ├── embedding.py │   │   │   ├── translation.py │   │   │   ├──
transcription.py │   │   │   └── pdf_generator.py │   │   └── utils/ │   │       ├── hashing.py │   │
├── encryption.py │   │       └── audit_logger.py │   ├── alembic/ │   ├── tests/ │   └── Dockerfile
├── frontend/ │   ├── app/ │   │   ├── page.tsx             -- homepage with India map + ticker │   │
└── constituency/[id]/   -- constituency dashboard │   └── components/ │       ├── IndiaMap.tsx
-- TopoJSON heat map │       ├── IssueLedger.tsx      -- top 10 issues with badges │       ├──
TrendBadge.tsx       -- Rising/Tatkal/Chronic/Stable │       └── QuestionLog.tsx      -- draft
questions ledger ├── docker-compose.yml ├── .env.example ├── SESSION_STATE.md            -- ALWAYS
update at end of session ├── DECISIONS.md                 -- log every non-obvious decision └──
README.md
```

SECTION 10 — BUILD ORDER & EXIT GATES

Do not skip steps. Do not reorder. Each week builds on the previous. The 8 Phase 1 exit gates must ALL be true before Phase 2 begins.

Weeks	Build	Definition of Done	Critical test
1-2	PostgreSQL + pgvector + all 11 migrations + 543 constituency seed data + FastAPI skeleton + Redis + Celery	alembic upgrade head clean; SELECT count(*) FROM constituencies = 543; GET /health = 200; Celery worker receives test task	docker-compose up; all checks pass
3-4	Intake Agent: WhatsApp webhook + Whisper + IndicTrans2 + Claude Haiku extraction + embedding storage + Aadhaar OTP + citizen acknowledgment	Submit Hindi voice note via WhatsApp → stored with all 8 fields populated → acknowledgment received within 60 seconds	End-to-end with real phone number
5	Clustering Agent: BERTopic + DBSCAN + velocity + badge assignment + Top 10 ledger	200 seeded issues → clusters make semantic sense; Tatkal badge triggers at correct threshold; Top 10 API returns correct data	pytest tests/test_clustering.py — all assertions pass
6	Question Hour Draft Agent: ministry mapping RAG + Rule 32 compliance + HARD GATE for unsourced claims	10 generated questions: all have 4 sub-parts, Rule citation, ≥2 sources, constituency evidence; hard gate blocks unsourced claims	Human review scores ≥8/10 'could be filed by an MP'
7	MP Brief Generator (PDF + WhatsApp) + public dashboard (map + ledger + question log)	PDF renders correctly; WhatsApp delivery confirmed to test number; dashboard loads on Chrome Android + Safari iOS	End-to-end brief generation and delivery test
8	Full integration test + load test + security audit prep + MP co-pilot onboarding	Full 8-step loop completes for 10 synthetic citizens across 3 constituencies; 1,000 issues injected in 60s → clustering completes in <15 min	pytest tests/test_full_loop.py — all 10 scenarios pass

10.1 Phase 1 Exit Gates — ALL 8 Must Be True

#	Gate	Verified by
1	1 parliamentary question derived from AgentSabha citizen cluster data filed in real Lok Sabha session	Lok Sabha bulletin / official session record with question number
2	Citizen whose cluster triggered the question received WhatsApp notification	Delivery receipt + citizen screenshot or verbal confirmation
3	1 national publication story on the co-pilot model with full loop documented	Published article URL
4	Public dashboard live showing real data for 3 constituencies	URL accessible without login; data is real, not seeded
5	100% fact-check pass rate on all MP briefs — zero unverified claims	agent_logs: zero fact_check_failed records
6	Third-party security audit completed; critical findings resolved	Security firm sign-off document
7	IT Rules 2021 + DPDP consent framework reviewed by legal counsel	Legal counsel memo or email

8	1 civil society organisation formally endorsed in writing	Written endorsement or press release

SECTION 11 — REVENUE MODEL & PROJECTIONS

11.1 The 4 Agents That Make the Most Money

Rank	Agent	Revenue event	Why it wins
#1	Election Intelligence Agent	₹5-50Cr per Lok Sabha cycle	Political parties pay crores for exit polls built on 10,000 samples. You have millions of verified, continuous, real-time constituency data points. No competition. This single event funds the entire platform.
#2	Vidhan Sabha Licensing	₹5-15Cr per state × 28 states	Once built for 543 Lok Sabha, replicating for 4,033 MLAs is a deployment project, not a rebuild. Near-100% gross margin.
#3	Financial Sector Risk Agent	₹20-150Cr/year	Insurance and banks price risk with 5-year-old census data. You have live constituency distress signals. That gap is worth hundreds of crores to insurers alone.
#4	Scheme Eligibility Agent	₹100Cr/year at scale	₹1.5 lakh crore in unclaimed entitlements annually. You already have the citizen profile. Matching them to schemes and taking a referral fee is the cleanest B2C model imaginable.

11.2 Revenue Projection Summary

Vertical	Year 1	Year 3	Year 5	Model
MP Intelligence (₹50K/mo)	₹2Cr	₹15Cr	₹40Cr	₹50K/month per MP subscription
Citizen Welfare (B2C)	₹50L	₹5Cr	₹25Cr	Freemium + referral fees
Government Intelligence (B2G)	₹2Cr	₹20Cr	₹80Cr	Annual state contracts
Corporate Intelligence (B2B)	₹1Cr	₹15Cr	₹60Cr	SaaS subscriptions
Financial Sector (B2B)	₹2Cr	₹25Cr	₹100Cr	Risk score API
Media & Research	₹50L	₹8Cr	₹25Cr	API + election cycle spikes
Platform & Licensing	₹0	₹10Cr	₹200Cr	Licensing + marketplace
TOTAL	~₹7.5Cr	~₹98Cr	~₹530Cr	Conservative estimate

> *Revenue sequencing: Phase 1 = ₹0 (trust only). Phase 2 = MP subscriptions + journalist API + YouTube. Phase 3 = government contracts + corporate SaaS + B2C welfare. Phase 4 = financial sector + election intelligence + platform licensing. Selling election or financial intelligence before civic trust is established is the fastest way to become the story instead of covering it.*

SECTION 12 — SESSION HANDOFF PROTOCOL

> *Every LLM session MUST start by reading this document. Every session MUST end by updating SESSION_STATE.md. This is what enables handoff without regressions.*

12.1 Start of Every Session — Do This First

1. Read Sections 1, 2, and 3 of this document to restore full context.
2. Read SESSION_STATE.md in repo root — understand what was last done.
3. Read DECISIONS.md last 20 lines — understand recent architectural choices.
4. Run: git log --oneline -10 — see what was committed.
5. Run: pytest tests/ — see current test status (no regressions before starting).
6. ONLY THEN write any code.

12.2 SESSION_STATE.md Template

```
# AgentSabha — Session State Last updated: [ISO datetime] Last developer: [name / LLM session] Current phase: Phase [1/2/3/4] Current week: Week [X] of 8  ## Completed this session - [task]: [DoD verified]  ## In progress (DO NOT restart from scratch) - [task]: [current state + next step]  ## Blocked - [task]: [blocker + what is needed]  ## Decisions made (add to DECISIONS.md) - [decision]: [reasoning]  ## Next session must start with - [specific task or test]  ## Known issues (bugs found, not yet fixed) - [issue description]
```

12.3 DECISIONS.md Template

```
## [Date] — [Decision title] Context: [why this decision was needed] Decision: [what was decided] Alternatives: [what else was considered] Reason: [why this was chosen] Consequences: [what this means for future work]
```

12.4 The One Question Before Any Code Change

> *Does this change violate any constraint in Section 2? If yes → stop and re-read Section 2 before proceeding.*

12.5 If Stuck or Unsure

- Re-read the relevant section of this document first.
- Check DECISIONS.md — the answer may already have been decided.
- If still unclear: write a comment # DECISION NEEDED: [question] and leave code incomplete. Document in SESSION_STATE.md under Blocked.
- NEVER guess a business rule, schema column, API behaviour, or agent behaviour. If it is not in this document, it has not been decided yet. Ask before implementing.

END OF MASTER BUILD PROMPT v3.0

47+ Agents · 543 Constituencies · ₹530Cr ARR · One Parliament

"We don't replace the MP. We hold them to a standard they have never been held to before."

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